

OPTIONAL TASK

Name of Student : Anuja Vilas Wankhade

Domain: Artificial intelligence and machine Learning (AIML)

College Name: Manav School of Polytechnic AKOLA

Branch : Computer Engineering

GitHub Repository Link: <https://github.com/anujaw193-star/python-project-collection>

❖ (Setup and Libraries)

```
1 import streamlit as st
2 import pandas as pd
3 import joblib
4
```

❖ (Loading Model and Preprocessing Objects)

```
# -----
# Load Model Files
# -----
try:
    model = joblib.load("models/model.pkl")
    scaler = joblib.load("models/scaler.pkl")
    encoded_columns = joblib.load("models/columns.pkl")
except Exception as e:
    st.error(f"Error loading model files:\n{e}")
    st.stop()
```

❖ (Page Configuration)

```
# -----
# Page Configuration
# -----
st.set_page_config(
    page_title="Ford Car Price Prediction",
    page_icon="🚗",
    layout="centered"
)
```

❖ (Title and Description)

```
# -----
# Title
# -----
st.title("🚗 Ford Car Price Prediction")
st.write("Enter the car details below and click **Predict Price**.")

st.divider()
```

❖ (Numerical Input Fields)

```

33 # -----
34 # User Inputs
35 # -----
36 year = st.number_input("Year", min_value=1990, max_value=2035, value=2018)
37
38 mileage = st.number_input("Mileage", min_value=0, value=25000)
39
40 tax = st.number_input("Tax", min_value=0, value=150)
41
42 mpg = st.number_input("MPG", min_value=0.0, value=55.4)
43
44 engineSize = st.number_input("Engine Size", min_value=0.5, value=1.5)
45
46 model_name = st.selectbox(
47     "Model",
48     [
49         "Fiesta",
50         "Focus",
51         "Kuga",
52         "Mondeo",
53         "EcoSport",
54         "Ka+",
55         "Puma",
56         "S-MAX",
57         "Galaxy",
58         "Edge",
59         "B-MAX",
60         "C-MAX",
61         "Grand C-MAX",
62         "Mustang",
63         "Tourneo Custom"
64     ]
65 )

```

❖ (Categorical Input using Dropdowns)

```

64     ]
65 )
66
67 transmission = st.selectbox(
68     "Transmission",
69     ["Manual", "Automatic", "Semi-Auto"]
70 )
71
72 fuelType = st.selectbox(
73     "Fuel Type",
74     ["Petrol", "Diesel", "Hybrid", "Electric", "Other"]
75 )
76

```

❖ (Text Input and Predict Button)

```

# -----
# Predict Button
# -----
if st.button("Predict Price"):
    try:
        (variable) input_df: DataFrame
        input_df = pd.DataFrame({
            "year": [year],
            "mileage": [mileage],
            "tax": [tax],
            "mpg": [mpg],
            "engineSize": [engineSize]
        })
    except:

```

❖ (Creating Input DataFrame & Encoding)

```
# Scale Numerical Features
scaled = scaler.transform(input_df)

scaled_df = pd.DataFrame(
    scaled,
    columns=["year", "mileage", "tax", "mpg", "engineSize"]
)

# One-Hot Encoding
cat_df = pd.DataFrame({
    f"model_{model_name}": [1],
    f"transmission_{transmission}": [1],
    f"fuelType_{fuelType}": [1]
})
```

❖ (Feature Scaling and Prediction)

```
# Combine
final_df = pd.concat([scaled_df, cat_df], axis=1)

# Match Training Columns
final_df = final_df.reindex(columns=encoded_columns, fill_value=0)

# Prediction
prediction = model.predict(final_df)

st.success(f"👉 Predicted Price: ₹ {prediction[0]:,.2f}")

except Exception as e:
    st.error(f"Prediction Error:\n{e}")
```

Output:



Ford Car Price Prediction

Enter the car details below and click Predict Price.

Year

2018 - +

Mileage

25000 - +

Tax

150 - +

MPG

55.40 - +

Engine Size

1.50 - +

Model

Fiesta

Fiesta

Focus

Kuga

Mondeo

EcoSport

Ka+

Puma

Transmission

Manual

Manual

Automatic

Semi-Auto

Fuel Type

Petrol

Petrol

Diesel

Hybrid

Electric

Other

Predict Price

Predicted Price: ₹ 15,256.03

Summary

In this project, a Streamlit frontend was developed for the Ford Car Price Prediction model. The application allows users to enter car details such as year, mileage, tax, MPG, engine size, model, transmission, and fuel type through a simple and user-friendly interface. The trained machine learning model, scaler, and encoded columns are loaded using Joblib. After preprocessing the input data, the model predicts the estimated price of the car and displays the result instantly. This project demonstrates the integration of a machine learning model with a web-based interface, making predictions easy and interactive for users.